

Michael Zuo

mikeyzuo2399@gmail.com | (513)-498-9821 | linkedin.com/in/mikey-zuo | github.com/Bimikel

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S./M.S. in Computer Science, B.S. in Mathematics – GPA: 4.0/4.0

May 2027

- **Relevant Coursework:** Data Structures, Honors Algorithms, Systems & Networks, Artificial Intelligence, Databases, Machine Learning, Computer Architecture, Deep Learning, Perception & Robotics, Real Analysis
- **Awards:** Y Combinator Startup School 2026, AIME 4x Qualifier, USA Physics Olympiad Semifinalist

EXPERIENCE

Meta

May 2026 – Present

Software Engineering Intern

Menlo Park, CA

- Developed a decision-tree analysis engine in Python and C++ that automatically identifies obsolete deployment gates, enabling large-scale cleanup across Meta's entire codebase used by **10K+ engineers**
- Implemented a rollout workflow in **Meta's code landing pipeline**, enabling engineers to apply recommended deployment gate changes during code submission for a platform serving **50K+ daily sessions**

Georgia Institute of Technology

Jan. 2026 – Present

Machine Learning Researcher

Atlanta, GA

- Developed **WeldNet**, a **deep learning framework** for applying dimensionality reduction techniques to transform high-dimensional sensor data into stable model representations using PyTorch
- Conducted empirical testing across multiple neural network architectures, analyzing performance through experiments with various differential equations and complex dynamical systems

College of Computing at Georgia Tech

Jan. 2025 – Present

Head Algorithms Teaching Assistant

Atlanta, GA

- Design and publish algorithms assignments and solutions for **300+ students**, aligning with course learning goals and strengthening understanding of graph theory, dynamic programming, and time complexity analysis
- Responded to **500+** student questions across EdStem, emails, and office hours with quick turnaround, improving clarity and accessibility of course support

Amazon

May 2025 – Aug. 2025

Software Development Engineer Intern

Bellevue, WA

- Cut business onboarding time by **90% (20 to 2 business days)** by building a CI/CD-driven workflow automation system with a self-service form in Java and TypeScript
- Automated deployment and integration pipelines with AWS Lambda and DynamoDB, replacing manual email processes with API-based verification

PROJECTS

Poker Strategy Evaluation Engine | *Python, FastAPI*

- Built a backend system to evaluate poker hands against over **1000** simulated opponent ranges per request with sub-**200ms** response times, exposed with a schema-validated FastAPI service
- Designed a reproducible decision pipeline that computes hand strength and recommendations, validated through unit tests and structured logging to support debuggable analysis

Pathfinder | *C++, Python*

- Implemented approximation algorithms for the Traveling Salesman Problem, including branch-and-bound, Christofides, and local search heuristics
- Generated solutions within **3%** of optimal on standard TSPLIB benchmarks while reducing runtime by over **90%** compared to exhaustive search

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, PHP, SQL, HTML/CSS, XML

Frameworks: React, Next.js, Node.js, Django, Flask, Spring Boot, GraphQL, REST APIs, Nginx, Android Studio

Databases: MySQL, PostgreSQL, MongoDB, Firebase

Cloud & Tools: AWS (Lambda, DynamoDB, CDK, EC2, S3, ECS, CloudWatch), Docker, Linux, Git/GitHub, CI/CD, JUnit, Jest

AI & ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, Machine Learning, Deep Learning